



COGNITION & MOOD

P21 (adamantane)

P21 (adamantane) – 10 mg per vial – 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

CHEMICAL NAME	P021; adamantylated CNTF-derived tetrapeptide derivative
SYNONYMS / ALIASES	P021; P21 adamantane; Ac-DGGL(Ada)G-NH2
CAS NUMBER	Not established in public registries — confirm against batch COA
MOLECULAR FORMULA	C30H54N6O5 (vendor-reported; confirm against batch COA)
MOLECULAR WEIGHT	Approximately 578 g/mol (confirm against batch COA)
AMINO ACID SEQUENCE	Ac-Asp-Gly-Gly-Leu-(adamantyl)Gly-NH2 — acetylated N-terminus, C-terminal amide, adamantylated C-terminal glycine
SOURCE	Synthetic (solid-phase peptide synthesis with adamantyl modification)

PHYSICAL PROPERTIES

APPEARANCE	Lyophilized white to off-white powder
PURITY SPECIFICATION	>=98% by HPLC (confirm against batch COA)
CONTAINER	10 mg per vial, sealed glass vial, rubber-stoppered, aluminum crimp

STORAGE & HANDLING

UNRECONSTITUTED STORAGE	-20 C, protect from light and moisture
RECONSTITUTED STORAGE	2-8 C, use within approximately 28 days
SUGGESTED RECONSTITUTION SOLVENT	Bacteriostatic water or sterile water (research use only)
SHELF LIFE (UNRECONSTITUTED, SEALED)	24 months at -20 C
STABILITY NOTES	Avoid repeated freeze-thaw cycles. Protect reconstituted solution from light.

STRUCTURAL NOTES

P021 is the adamantylated derivative of the CNTF 148-151 core motif (DGGL). The adamantane group and the N-terminal acetyl / C-terminal amide capping were introduced to increase metabolic stability and lipophilicity relative to the unmodified peptide. This is the form characterised in the published literature cited below.

SELECTED PEER-REVIEWED REFERENCES

1. Kazim S.F. et al. — *Disease modifying effect of chronic oral treatment with a neurotrophic peptidergic compound in a triple transgenic mouse model of Alzheimer's disease* — *Neurobiology of Disease*, 2014;71:110-130 — DOI: 10.1016/j.nbd.2014.07.001
2. Kazim S.F., Iqbal K. — *Neurotrophic factor small-molecule mimetics mediated neuroregeneration and synaptic repair: emerging therapeutic modality for Alzheimer's disease* — *Molecular Neurodegeneration*, 2016;11(1):50 — DOI: 10.1186/s13024-016-0119-y

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

COMPLIANCE DISCLAIMER

This product is sold by Aligned Aminos for laboratory research use only (RUO). It is not a drug, dietary supplement, food, cosmetic, or medical device. It has not been approved by the FDA for any therapeutic, diagnostic, or preventative use. It is not intended for use in humans or animals, and may not be administered, ingested, applied, or otherwise introduced into any living organism. Purchasers represent that they are qualified researchers or laboratory professionals and assume full responsibility for compliance with all applicable laws and institutional review requirements governing the handling of research chemicals.

Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.